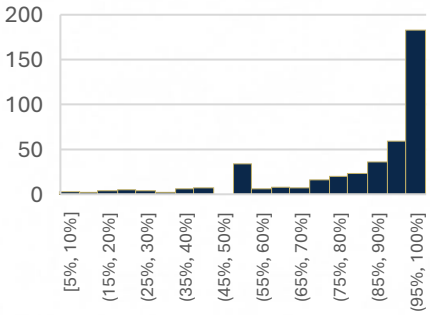
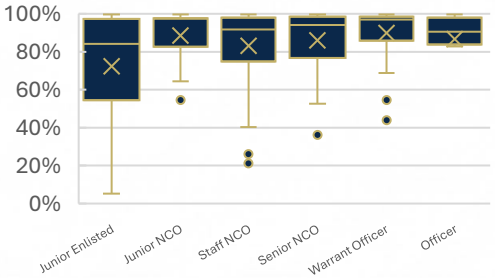
Artificial Intelligence in HUMINT Operations: Assessing Feasibility via Cognitive Load Theory

Presenter: Jonathan Capers, US Army
National Intelligence University

FEASIBILITY DISTRIBUTION

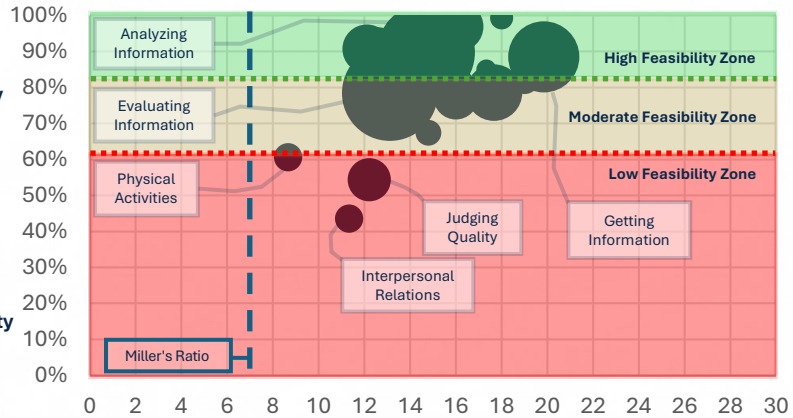


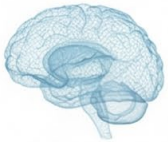


- 190 of 426 Critical Tasks > 95% feasible for AI Integration
- Officer tasks are 9% more feasible than Enlisted counterparts' tasks
- Risk Index (*RI*) is the most critical factor to evaluate Feasibility (*F*)

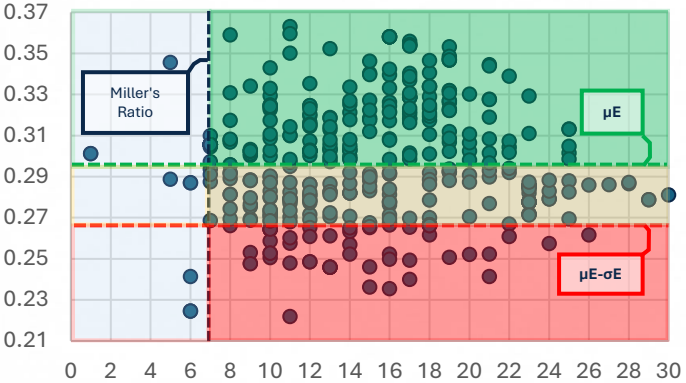
COGNITIVE OFFLOADING MAP

- Highly analytic, data-driven tasks massively outweigh interpersonal and subjective tasks
- Inextricably-human tasks demonstrate lowest feasibility



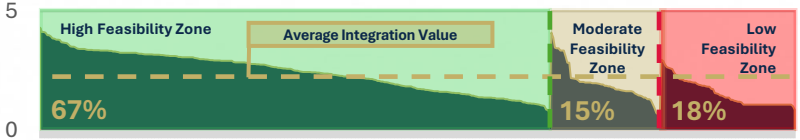


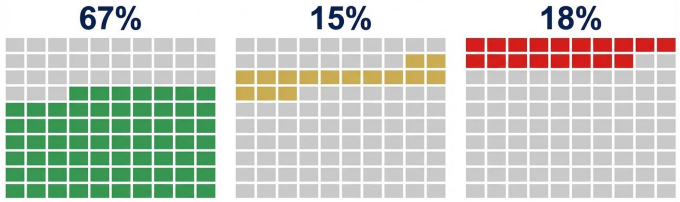
CAPACITY VS. CAPABILITY



- 182 tasks (*L* > 7 and *E* > 29.68%) are optimal for AI offloading
- 140 tasks (*L* > 7 and 26.91% < *E* < 29.68%) require human-in-the-loop oversight
- 63 tasks (*L* > 7 and *E* < 26.91%) demonstrate low AI capability and high potential failure

INTEGRATION RECOMMENDATIONS





High Feasibility
Tasks that have a Feasibility (*F*) value above 83.1% (μ) are prime targets for AI Integration

Moderate Feasibility
Tasks that have a Feasibility (*F*) value less than 83.1% (μ*F*) but more than 61.2% (μ*F*-σ*F*)

Low Feasibility
Tasks that have a Feasibility (*F*) value less than 61.2% (μ*F*-σ*F*)

NIU NATIONAL INTELLIGENCE UNIVERSITY
The IC's Bridge to the Future